

Roman Jahandideh

Human–Computer Interaction · Bioadaptive XR · Embodied & Collaborative Systems

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romanjahandideh.com · github.com/RomanJahandideh portfolio



portfolio

RESEARCH PROFILE

I design and study digital environments: the generative tools, virtual spaces, and responsive interfaces that increasingly sit between people and the physical world they are trying to understand, build, or inhabit. My work moves across that territory. Right now it includes a game world in Unity driven by live heart-rate data, an ambient cue that lets remote teammates sense each other without reading numbers, and earlier research on how people perceive depth and distance inside head-mounted virtual environments. I came to this through architecture and landscape design, and it is where I first started asking the question that still drives everything I build: how can digital environments, whether a generative design tool, a virtual space, or a responsive interface, be designed to optimize the real, physical spaces we live in and the lives we live inside them? That question runs in a few directions depending on the project: using generative, AI-assisted design to make complex spatial and environmental information legible and usable; understanding how people perceive, move through, and physically respond to 3D space; and building systems that treat the body itself as a source of spatial and environmental information.

RESEARCH INTERESTS

Generative & AI-assisted design for the built environment · Physiologically adaptive & biofeedback-driven systems · Embodied interaction in virtual & game environments · Human-centered empirical user studies

EDUCATION

Master's in Interactive Arts and Technology, Simon Fraser University (Surrey, BC) 2025 – present

- My second master's, on bioadaptive game worlds, wearable sensing and self-regulation. Graduating this semester.

Master's in Landscape Architecture Design, Istanbul Technical University 2020 – 2024

- Thesis work in the Virtual Landscape / Interface X Lab on depth perception in head-mounted displays.

Bachelor's in Architecture Design, Tabriz Islamic Art University (Iran) 2011 – 2016

Relevant graduate coursework: Human–Computer Interfaces, Serious Games, Research Design, Scientific Research Ethics and Seminar, Contemporary Art.

PUBLICATIONS AND MANUSCRIPTS

Jahandideh, R., et al. (2023). *Creating Immersive Virtual Landscapes: A User-Centred Approach to Enhance Depth Perception in Head-Mounted Displays*. eCAADe 2023: Digital Design Reconsidered, Vol. 1, p. 149. doi:10.52842/conf.ecaade.2023.1.149

Jahandideh, R., et al. (2024). *Optimizing Eye-Tracking Locomotion in Virtual Environments: The Effects of Colour, Luminance, Light Intensity and Movement Speed*. Full paper accepted, CAADRIA 2024.

Jahandideh, R. (2026). *Reading a Remote Teammate's Body: A Body-Worn Sensing Interface for Group Coordination in a Distributed Interactive Space*. Under review, ACM. Icon-legibility study (n = 36) and live three-player cooperative play (12 teams).

Jahandideh, R. (2026). *Embedding Physiological Responsiveness into Game World Dynamics: Designing Bioadaptive Game Worlds for Embodied Self-Regulation*. Submitted to ACM SIGGRAPH 2026. Within-subjects study, n = 24.

In preparation: *The Illusion of Distance* (object transparency and egocentric distance in VR), *Lights Out, Way On* (eye-tracking locomotion), and *The Human Touch in AI Architecture*.

RESEARCH EXPERIENCE

Graduate Researcher, iViz Lab, School of Interactive Arts + Technology, SFU 2025 – present

- BioTerrarium, a survival game in Unity 6 that listens to heart rate and HRV from an Apple Watch. Fog thickens and plants overgrow when the player tenses up, and the world clears once they settle. Prototype at biofeedbackexperiment.netlify.app
- Built the sensing pipeline: HealthKit into a Swift bridge, then Bluetooth and a WebSocket into Unity, under 200 ms end to end, with four arousal zones set from each player's own resting baseline.
- Ran the study behind it: 24 participants, two sessions each a day apart, HR and HRV logged alongside PANAS, FSS and MAAS. Paired t-tests and correlation for the numbers, thematic analysis for the interviews.
- A second project: a glowing icon in the HUD carrying a teammate's heart rate and its variability, with no numbers and no emotion words. 36 people read it online first, then 12 three-player teams played with it.

Researcher, Interface X Lab, Istanbul Technical University 2020 – 2024

- VR experiments in Unreal Engine with head-mounted displays, on how people judge depth and distance when transparency, colour and light levels change.
- Ran the eye-tracking locomotion studies behind two of the papers above, handled visuals for seven of the lab's projects, and taught undergraduates how a research paper is put together.

VR and UX Developer, Computer and Technology Lab, Tabriz Islamic Art University 2014 – 2016

- VR simulations that let design students walk through their own work and find the mistakes, plus modelling, animation and lighting for teaching material.

TEACHING

- Teaching Assistant, IAT 102 Graphic Design, SFU, Sept 2026 to present. Studio and lab sessions, critique, grading.
- Teaching Assistant, IAT 343 3D Animation, SFU, May 2025 to present. Three semesters, 40-plus students, mostly Maya and Cinema 4D. Teams bring me broken rigs and render settings, and I walk them through a short from storyboard to final cut.
- Teaching Assistant, IAT 313 Narrative and New Media, SFU, May 2025 to Jan 2026. A writing-intensive course: weekly discussion, feedback on interactive work in Unity and Unreal, and a lot of time on drafts.
- Before SFU: Photoshop in Urmia from 2015 to 2019, then 3ds Max and architectural rendering in groups of three to five until 2020. Workshops at Istanbul Technical University, English tutoring on Cambly.

UNIVERSITY SERVICE

- Senator, SFU Senate, for the Faculty of Communication, Art and Technology, March to June 2025. Teaching and research policy, course development and curriculum review.
- Senate Graduate Studies Committee, SFU, March 2025 to present. New programs, course changes, and thesis and supervision guidelines come through for review.

AWARDS AND CREATIVE PRACTICE

- Best Dissertation Award, Tabriz Art University, 2016. Second prize in the university's landscape design competition, 2013. Sixth prize, Sustainable Cabin for Natural Disasters, Memar News, 2018. Best abstract photography, Tabriz Art Festival, 2018. NODET (Iran), 2004 to 2011.
- Two digital art collections released independently on OpenSea and Foundation, 2020 to 2022.

TECHNICAL SKILLS

Programming: Python (two University of Michigan certificates), C# in Unity, Swift for the HealthKit and Bluetooth bridge. JavaScript with D3.js, Three.js and WebGL for web prototypes and visualization, plus HTML and CSS. Visual scripting in Unreal Blueprints and Grasshopper.

Sensing and data: Apple Watch and HealthKit, Bluetooth Low Energy, WebSocket streaming, networked multiplayer prototypes, heart rate and HRV (RMSSD), baseline normalization and zone mapping, feedback loops under 200 ms.

Real-time 3D: Unity 6, Unreal Engine, Autodesk Maya, Blender, Cinema 4D, 3ds Max, Rhinoceros + Grasshopper.

Research methods: Within-subjects and counterbalanced designs, research-through-design, PANAS, FSS and MAAS, t-tests and correlation, semi-structured interviews, thematic analysis, university ethics approval.

Languages: Persian and Azerbaijani (native), Turkish (advanced), English (research, publication and teaching).

ACADEMIC WRITING AND RESEARCH TOOLS

Writing: Overleaf and LaTeX with the ACM acmart template and BibTeX. Word and Google Docs when co-authors prefer them. Camera-ready formatting of figures, tables and references.

References: Zotero for group libraries, PDF annotation and BibTeX export, switching between ACM, APA and IEEE styles. Google Scholar and the ACM Digital Library for reviews; anonymized versions and supplementary files for submission.

REFERENCES

Dr. Steve DiPaola, Professor, School of Interactive Arts + Technology, Simon Fraser University

sdipaola@sfu.ca · Course supervisor, IAT 343 Animation (Teaching Assistant, 2025 – present)

Susan Clements-Vivian, Associate Director and University Lecturer, School of Interactive Arts + Technology, Simon Fraser University

sclement@sfu.ca · Course supervisor, IAT 313 Narrative & New Media (Teaching Assistant, 2025 – 2026)

Dr. Ikhwan Kim, Assistant Professor, Department of Landscape Architecture, Istanbul Technical University

kimi@itu.edu.tr · iikim3@gmail.com · Research supervisor, virtual landscape and VR lab, ITU (2020 – 2024)